Multiclass Prediction of Osteoporosis Status Using Clinical and Socioeconomic Factors in NHANES 2017-2018

Ruoxi Yang

Abstract

Background: Osteoporosis and osteopenia are common conditions associated with increased fracture risk and substantial public health burden. Bone mineral density (BMD)–based T-scores are the clinical standard for osteoporosis classification, yet the extent to which demographic, anthropometric, lifestyle, and socioeconomic factors predict osteoporosis status remains an active area of investigation.

Methods: A cross-sectional analysis was conducted using adult participants from the 2017-2018 National Health and Nutrition Examination Survey (NHANES). Femoral neck T-scores were calculated using a standard young adult female reference mean and standard deviation, and participants were classified as normal, osteopenic, or osteoporotic according to World Health Organization thresholds. Multiclass prediction models were developed using multinomial logistic regression and random forest classifiers. Model performance was evaluated using five-fold cross-validation, with assessment based on confusion matrices, macro-F1 score, and balanced accuracy. The predictive contribution of social determinants of health (SDoH) was evaluated by comparing full models with reduced models excluding education and income-to-poverty ratio. Missing data were addressed using complete-case analysis.

Results: The prevalence of normal bone density, osteopenia, and osteoporosis was approximately 52.6%, 40.7%, and 6.7%, respectively, with marked differences observed across sex and race/ethnicity groups. Multinomial logistic regression achieved higher macro-F1 scores and balanced accuracy than random forest models. Exclusion of SDoH variables resulted in modest improvements in predictive performance across both modeling approaches. Feature importance analysis based on SHAP values from the random forest model identified age, body mass index, waist circumference, sex, and menopausal status as the most influential predictors.

Conclusions: Multiclass prediction of osteoporosis status using routinely collected NHANES variables is feasible, with interpretable regression models demonstrating performance comparable to machine learning approaches. In this dataset, socioeconomic variables provided limited incremental predictive value beyond core biological and anthropometric factors.

1 Introduction

Osteoporosis is a prevalent skeletal disorder characterized by reduced bone mass and microarchitectural deterioration, leading to increased fracture risk, morbidity, and healthcare burden among older adults [1]. In the United States, osteoporosis and low bone mass affect tens of millions of individuals, with particularly high prevalence among postmenopausal women and older adults [2]. Osteoporotic fractures are associated with substantial functional decline, increased mortality, and rising medical costs, underscoring the importance of early identification and risk stratification [3].

Dual-energy X-ray absorptiometry (DXA)–derived bone mineral density (BMD) remains the clinical gold standard for osteoporosis assessment. The World Health Organization (WHO) defines osteoporosis and osteopenia using T-score thresholds based on deviation from a young adult reference population. While these thresholds are widely applied in clinical practice, population-level characterization of osteoporosis status and its predictors remains an important public health objective, particularly using nationally representative datasets.

The National Health and Nutrition Examination Survey (NHANES) [4] provides a unique opportunity to examine osteoporosis prevalence and risk factors in the U.S. population using standardized DXA measurements alongside detailed demographic, anthropometric, lifestyle, health history, and socioeconomic data. Prior studies have identified age, sex, body mass index, menopausal status, and comorbid conditions as key correlates of low bone density. However, the relative predictive value of social determinants of health (SDoH), such as education and income, remains less clearly defined in multivariable prediction settings.

In recent years, machine learning approaches have been increasingly applied to clinical risk prediction, offering potential advantages in capturing nonlinear relationships and complex interactions. Nevertheless, the incremental benefit of such methods over traditional, interpretable statistical models in osteoporosis classification remains uncertain, particularly when restricted to routinely collected clinical and socioeconomic variables. Moreover, rigorous evaluation of multiclass prediction performance using appropriate metrics beyond overall accuracy is often lacking.

The present analysis aims to characterize osteoporosis status in older U.S. adults using NHANES 2017-2018 data and to evaluate multiclass prediction of normal bone density, osteopenia, and osteoporosis. Predictive performance of multinomial logistic regression and random forest models is compared using cross-validation, with emphasis on macro-F1 score and balanced accuracy. Additionally, the contribution of social determinants of health to predictive performance is quantified by comparing full models with reduced models excluding education and income-related variables.

2 Methods

This study used data from the 2017–2018 cycle of the National Health and Nutrition Examination Survey (NHANES) to examine demographic, clinical, and behavioral determinants of bone health and to develop predictive models of osteoporosis status. Participants aged 50 years and older with available dual-energy X-ray absorptiometry (DXA) bone mineral density (BMD) measurements were included. Predictor variables included anthropometrics, sociodemographic characteristics, lifestyle behaviors, clinical comorbidities, and reproductive factors. Missing data were addressed using appropriate imputation procedures. Multinomial logistic regression and random forest models were trained to classify participants into normal BMD, osteopenia, or osteoporosis

groups, with performance assessed using cross-validated accuracy, macro-F1, and balanced accuracy metrics. Model interpretability was evaluated using SHAP values to quantify the relative contribution of individual predictors.

2.1 Study Population

A cross-sectional analysis was conducted using data from the 2017-2018 National Health and Nutrition Examination Survey (NHANES), a nationally representative survey of the noninstitutionalized U.S. population. Participants aged 50 years and older were eligible for inclusion. Individuals with missing femoral neck bone mineral density (BMD) measurements were excluded. The final analytic cohort consisted of participants with complete data for outcome definition and model predictors.

2.2 Predictor Variables

Predictor variables were selected from demographic, anthropometric, lifestyle, health history, and socioeconomic domains based on prior literature and data availability. Demographic variables included age, sex, and race/ethnicity. Anthropometric measures included body mass index, body weight, height, and waist circumference. Lifestyle factors included smoking status, alcohol use, and physical activity. Health history variables included menopausal status, fracture history, rheumatoid arthritis, thyroid disease, hormone replacement therapy use, and pregnancy status where applicable. Social determinants of health (SDoH) were represented by educational attainment and income-to-poverty ratio. To prevent data leakage, no DXA-derived bone mineral density measures were included as predictors.

2.3 Bone Mineral Density and Outcome Definition

Femoral neck bone mineral density (BMD) was measured using dual-energy X-ray absorptiometry (DXA). T-scores were calculated using a standard young adult female reference population, consistent with World Health Organization (WHO) recommendations. The reference mean BMD was 0.858 g/cm^2 with a standard deviation of 0.120 g/cm^2 [5]. The femoral neck T-score was computed as:

$$\text{T-score} = \frac{\text{BMD}_{\text{participant}} - \mu_{\text{reference}}}{\sigma_{\text{reference}}},$$

where $\text{BMD}_{\text{participant}}$ denotes the observed femoral neck BMD, $\mu_{\text{reference}}$ denotes the reference mean BMD, and $\sigma_{\text{reference}}$ denotes the reference standard deviation. Participants were classified as having normal bone density (T-score ≥ -1.0), osteopenia ($-2.5 <$ T-score < -1.0), or osteoporosis (T-score ≤ -2.5) [6].

2.4 Missing Data Handling

Missingness was assessed for all candidate predictors. The largest proportions of missing data were observed for income-to-poverty ratio and fracture history. Given the moderate level of missingness and the modeling focus on prediction rather than causal inference, a complete-case analysis

was employed. The final analytic sample sizes for modeling were reported separately for full and reduced predictor sets.

2.5 Statistical Modeling

Multiclass prediction models were developed to classify participants into normal bone density, osteopenia, or osteoporosis categories. Two modeling approaches were implemented: multinomial logistic regression as an interpretable baseline model and random forest classification as a machine learning comparator [7]. Five-fold cross-validation was used to evaluate model performance. Performance metrics included confusion matrices, macro-F1 score, balanced accuracy, and overall accuracy [8]. To evaluate the contribution of social determinants of health, model performance was compared between full models including all predictors and reduced models excluding education and income-to-poverty ratio.

2.6 Model Interpretation

Feature importance for the random forest model was assessed using SHAP (Shapley Additive Explanations) values [9]. SHAP values were estimated for each outcome class using predicted class probabilities, and importance was summarized as the mean absolute SHAP value averaged across classes. This approach quantified the relative contribution of each predictor to multiclass classification performance.

3 Results

The results indicate clear demographic, behavioral, and clinical differences across osteoporosis categories. Osteoporosis was predominantly observed among older participants and women, with substantial variation across racial and socioeconomic groups. Individuals with osteoporosis consistently exhibited lower body weight, BMI, and bone density metrics, along with higher prevalence of selected clinical conditions. Predictive models achieved moderate discrimination between normal BMD, osteopenia, and osteoporosis, with only modest impact from incorporating social determinants of health. Feature interpretation analyses further demonstrated that anthropometric measures, age, and demographic characteristics were the primary drivers of model predictions.

3.1 Prevalence of osteoporosis categories

Figure 1 illustrates the prevalence of osteoporosis categories overall and stratified by gender and race/ethnicity. Overall, 52.6% of participants had normal bone mineral density (BMD), while 40.7% had osteopenia and 6.7% had osteoporosis. Substantial gender differences were observed. Among men, 65.1% had normal BMD, 32.5% had osteopenia, and only 2.4% had osteoporosis. In contrast, among women, the prevalence of normal BMD was considerably lower (38.9%), whereas the prevalence of osteopenia (49.7%) and osteoporosis (11.4%) was markedly higher. Marked variation was also noted across race/ethnicity. The prevalence of normal BMD ranged from 37.1% in Hispanic participants to 69.9% in Black participants, whereas osteopenia prevalence ranged from 28.4% in Black participants to 51.2% in Asian participants. Osteoporosis prevalence was highest

among White (10.2%) and multi-racial (8.0%) participants, followed by Hispanic (4.6%) and Asian (8.7%) participants, and was lowest among Black participants (1.7%). Error bars indicate 95% confidence intervals.

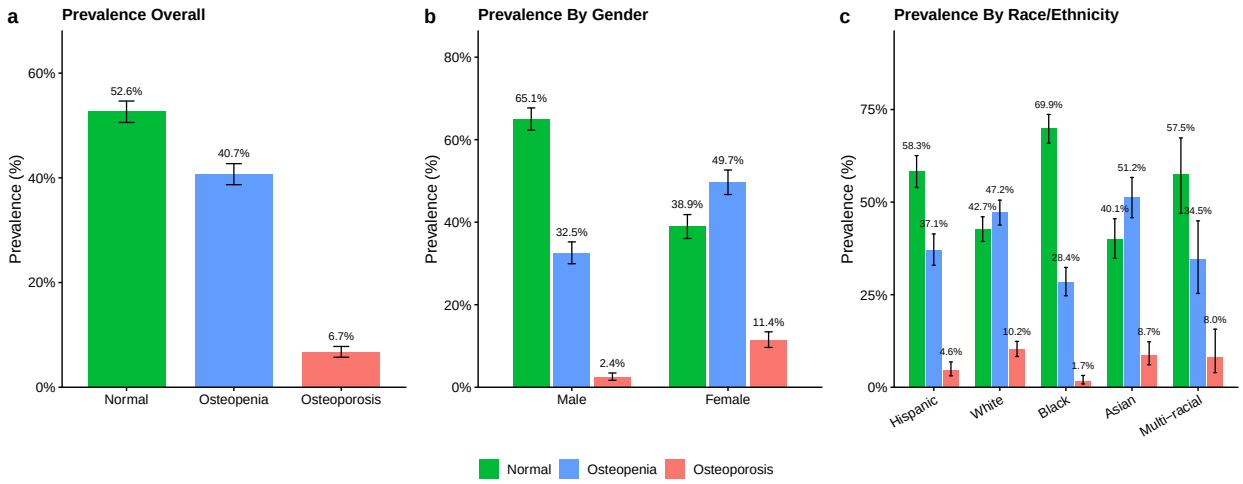


Figure 1: Prevalence of osteoporosis status overall and stratified by gender and race/ethnicity (NHANES 2017–2018, age ≥ 50). Error bars indicate 95% confidence intervals.

3.2 Baseline characteristics by osteoporosis status

Baseline characteristics by osteoporosis status are summarized in **Table 1**. Participants with osteoporosis were older on average compared with those with osteopenia and normal BMD (mean age 71.3 vs. 66.0 and 62.8 years, $p < 0.001$). Women constituted the majority of osteopenia and osteoporosis groups, whereas men were more commonly represented in the normal BMD group ($p < 0.001$). Racial patterns were also evident, with a greater proportion of White participants in the osteoporosis group and a greater proportion of Black participants in the normal group ($p < 0.001$). Socioeconomic indicators suggested slightly lower income levels (INDFMPIR) and educational attainment among individuals with osteoporosis, although differences were modest.

Individuals with osteoporosis were more likely to be physically inactive and less likely to report alcohol use ($p < 0.001$). Higher prevalence of thyroid disease and prior fracture was observed among participants with osteoporosis, whereas rheumatoid arthritis prevalence was similar across groups. As expected, anthropometric measures showed strong gradients across categories: weight, height, BMI, and waist circumference progressively decreased from normal to osteopenia to osteoporosis ($p < 0.001$). Bone density measures (femoral neck, total femur, trochanter, and lumbar spine BMD) demonstrated consistent and pronounced reductions across osteopenia and osteoporosis groups ($p < 0.001$). Among women, menopausal status, HRT use, and pregnancy history differed significantly across groups, with post-menopausal women more likely to have osteoporosis. Overall, these patterns highlight strong associations between osteoporosis status and age, sex, body composition, reproductive factors, and selected clinical conditions.

Table 1: Baseline characteristics of participants stratified by osteoporosis status (Normal, Osteopenia, Osteoporosis), NHANES 2017–2018.

Variable	Bone mineral density status			p-value[2]
	Normal ($N = 1203$)[1]	Osteopenia ($N = 930$)[1]	Osteoporosis ($N = 153$)[1]	
RIDAGEYR	62.84 (8.47)	65.98 (9.09)	71.31 (8.47)	<0.001
Gender				<0.001
Male	780 (65%)	390 (42%)	29 (19%)	
Female	423 (35%)	540 (58%)	124 (81%)	
Race				<0.001
Hispanic	291 (24%)	185 (20%)	23 (15%)	
White	361 (30%)	399 (43%)	86 (56%)	
Black	372 (31%)	151 (16%)	9 (5.9%)	
Asian	129 (11%)	165 (18%)	28 (18%)	
Multi-racial	50 (4.2%)	30 (3.2%)	7 (4.6%)	
Education				0.20
Less than HS	249 (21%)	189 (20%)	34 (22%)	
High school	289 (24%)	231 (25%)	52 (34%)	
Some college/AA	371 (31%)	276 (30%)	37 (24%)	
College graduate	290 (24%)	233 (25%)	30 (20%)	
Advanced degree	4.0 (0.3%)	1.0 (0.1%)	0 (0%)	
INDFMPIR	2.73 (1.60)	2.65 (1.60)	2.44 (1.50)	0.20
Physical activity, n (%)				<0.001
Inactive	336 (28%)	290 (31%)	71 (46%)	
Active	867 (72%)	640 (69%)	82 (54%)	
Alcohol	639 (56%)	405 (46%)	46 (30%)	<0.001
Smoking				<0.001
Never	598 (50%)	544 (68%)	90 (59%)	
Former	208 (17%)	120 (13%)	24 (16%)	
Current	397 (33%)	266 (29%)	39 (25%)	
Menopause				<0.001
Pre/Peri-menopausal	969 (81%)	577 (62%)	80 (52%)	
Post-menopausal	234 (19%)	353 (38%)	73 (48%)	
FractureHx	14 (1.3%)	16 (2.0%)	16 (12%)	<0.001
RheumatoidArthritis	105 (8.7%)	87 (9.4%)	15 (9.8%)	0.8
ThyroidDisease	109 (9.1%)	141 (15%)	34 (23%)	<0.001
BMXWT	87.28 (19.24)	73.05 (15.50)	63.59 (15.88)	<0.001
BMXHT	168.53 (9.51)	163.21 (9.50)	157.86 (8.17)	< 0.001
BMXBMI	30.70 (6.11)	27.40 (5.23)	25.42 (5.48)	<0.001
BMXWAIST	105.21 (14.11)	97.57 (13.28)	93.22 (14.97)	<0.001
DXOFBMD	1.03 (0.12)	0.83 (0.09)	0.64(0.08)	<0.001

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Variable	Bone mineral density status			p-value[2]
	Normal ($N = 1203$)[1]	Osteopenia ($N = 930$)[1]	Osteoporosis ($N = 153$)[1]	
DXXNKBMD	0.86 (0.10)	0.66 (0.05)	0.50 (0.06)	<0.001
DXXTRBMD	0.78 (0.11)	0.62 (0.08)	0.49 (0.08)	<0.001
DXXINBMD	1.22 (0.14)	0.99 (0.12)	0.77 (0.12)	<0.001
calcium	1.137 (95%)	861 (93%)	131 (86%)	< 0.001
vitamin	1.166 (97%)	896 (96%)	147 (96%)	< 0.7
HRT	102 (8.5%)	146 (16%)	33 (22%)	<0.001
Pregnancy	375 (31%)	463 (50%)	107 (70%)	<0.001

[1] Mean (SD); n (%).

[2] Kruskal–Wallis rank sum test; Pearson's Chi-squared test.

3.3 Model performance and SDoH comparison

Table 2 summarizes cross-validated performance for multiclass prediction of osteoporosis status. Overall, model performance was moderate across approaches, with accuracies ranging from 0.617 to 0.646. The multinomial logistic regression models achieved the highest overall accuracy, with performance being similar between the full model (0.644 accuracy, macro-F1 = 0.529, balanced accuracy = 0.509) and the model excluding SDoH variables (0.646 accuracy, macro-F1 = 0.538, balanced accuracy = 0.517). Random forest models performed slightly worse, with accuracies of 0.617 (macro-F1 = 0.485, balanced accuracy = 0.470) for the full model and 0.626 (macro-F1 = 0.504, balanced accuracy = 0.484) when SDoH variables were excluded. Across all modeling strategies, the inclusion of SDoH features did not substantially improve predictive performance, suggesting that clinical and demographic variables captured most of the discriminative information for osteopenia/osteoporosis classification.

Table 2: Cross-validated performance for multiclass prediction of osteoporosis status.

Model	Accuracy	Macro-F1	Balanced Accuracy
Multinomial LR (Full)	0.644	0.529	0.509
Multinomial LR (No SDoH)	0.646	0.538	0.517
RF (Full)	0.617	0.485	0.470
RF (No SDoH)	0.626	0.504	0.484

The confusion matrices in **Table 3** indicate that models excluding SDoH predictors generally perform better than those including them. Both multinomial logistic regression and random forest models show higher correct classification counts in the "No SDoH" setting, particularly for the Normal and Osteoporosis categories. In contrast, when SDoH variables are added, misclassification increases, especially with more Normal cases being classified as Osteopenia and more Osteoporosis cases being misclassified as Osteopenia. Comparing the two algorithms, random forest consistently

performs slightly better than multinomial logistic regression across both settings, but the overall pattern remains the same: removing SDoH predictors leads to clearer class separation and fewer classification errors.

Table 3: Confusion matrices for multinomial logistic regression and random forest models with and without SDoH predictors.

(a) Multinomial LR (Full)

	Normal	Osteopenia	Osteoporosis
Normal	628	221	4
Osteopenia	255	413	14
Osteoporosis	10	82	21

(b) Multinomial LR (No SDoH)

	Normal	Osteopenia	Osteoporosis
Normal	748	240	7
Osteopenia	304	450	19
Osteoporosis	14	87	28

(c) Random Forest (Full)

	Normal	Osteopenia	Osteoporosis
Normal	619	231	3
Osteopenia	292	383	7
Osteoporosis	21	78	14

(d) Random Forest (No SDoH)

	Normal	Osteopenia	Osteoporosis
Normal	732	259	4
Osteopenia	330	435	8
Osteoporosis	20	89	20

3.4 Feature importance (Random Forest SHAP)

Figure 2 shows the relative contribution of individual predictors to the Random Forest model [10] using mean absolute SHAP values. Anthropometric measures were the most influential predictors, with body weight (BMXWT), height (BMXHT), and BMI (BMXBMI) showing the highest SHAP importance, followed by age (RIDAGEYR) and waist circumference (BMXWAIST). Demographic characteristics, including gender, race/ethnicity, and socioeconomic status (INDFMPIR), also contributed meaningfully to model discrimination. Reproductive and hormonal factors such as menopause and pregnancy status exhibited moderate importance, while lifestyle variables including alcohol use, smoking, and physical activity showed comparatively smaller effects. Clinical comorbidities such as thyroid disease, rheumatoid arthritis, and prior fracture contributed minimally relative to anthropometric and demographic predictors. These results suggest that body composition–related measures remain the dominant determinants in distinguishing osteoporosis categories within the model.

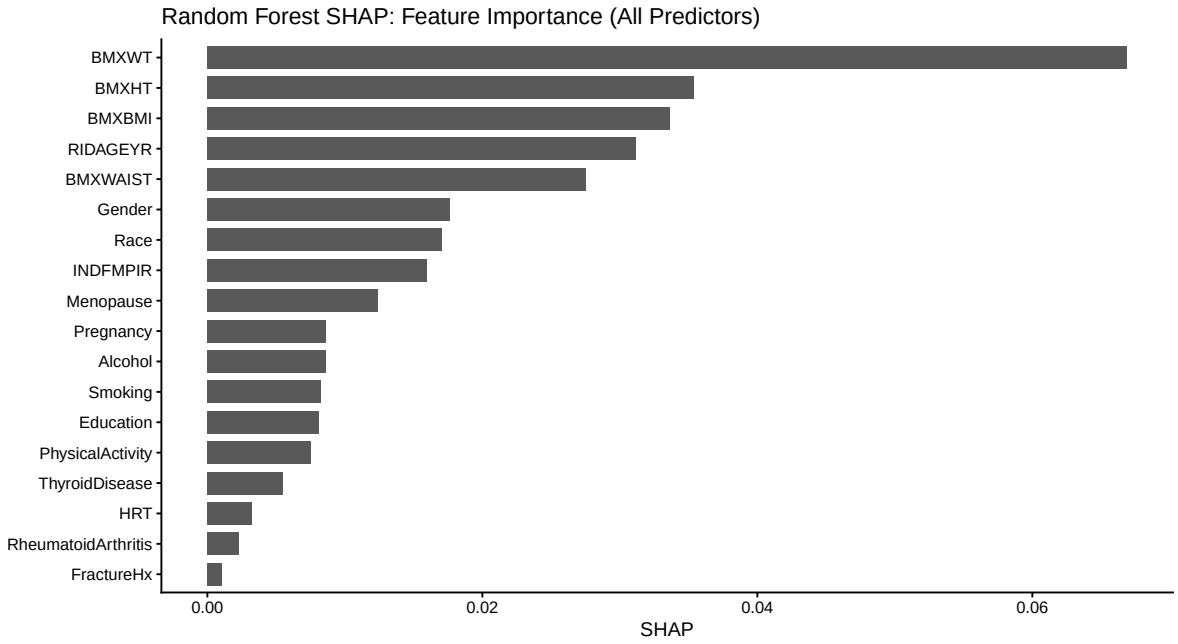


Figure 2: Random Forest feature importance using mean absolute SHAP values averaged across class probabilities.

4 Discussion

This analysis demonstrates that multiclass prediction of osteoporosis status using routinely collected NHANES variables is feasible, with interpretable statistical models performing comparably to machine learning approaches. Multinomial logistic regression consistently outperformed random forest models in terms of macro-F1 score and balanced accuracy, highlighting the value of parsimonious, interpretable models in settings with moderate sample sizes and imbalanced outcome classes.

The observed prevalence patterns by sex and race/ethnicity are consistent with prior epidemiologic studies, reinforcing known disparities in bone health among older adults. The reduced sensitivity for osteoporosis across models likely reflects both the lower prevalence of osteoporosis and overlapping risk profiles with osteopenia, which poses challenges for multiclass classification.

The comparison of full and reduced models suggests that social determinants of health contributed limited incremental predictive value beyond core biological and anthropometric predictors. Removal of education and income variables resulted in modest performance improvements, likely due to reduced model complexity and decreased noise. These findings do not diminish the importance of socioeconomic factors in osteoporosis risk but suggest that their effects may be mediated through or correlated with other measured predictors in this dataset.

Several limitations should be noted. The cross-sectional design precludes assessment of temporal relationships or fracture outcomes. Complete-case analysis may have introduced selection bias, although missingness was moderate for most predictors. Additionally, model performance may be constrained by class imbalance and the absence of longitudinal risk factors.

Future work may explore alternative modeling strategies, such as class-weighted algorithms or

ordinal classification frameworks, and evaluate prediction of clinically relevant outcomes such as incident fractures. Incorporation of longitudinal data and external validation may further enhance model generalizability and clinical utility.

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